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BMEN 309: PROJECT (RESEARCH-DESIGN)

PROJECT TITLE:

DESIGN OF A TAMPON-SIZED IMAGING DEVICE FOR CERVICAL EXAMINATION

,

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Content

1. Introduction 1
2. The Vaginal Speculum 2
   2.1 History of the Vaginal Speculum 2
   2.2 Common Materials Used in Design of Vaginal Speculum 3
   2.3 Patents Related to the Vaginal Speculum 4
3. Product Analysis 6
   3.1 Common Vaginal Speculum Brands 7
   3.2 Important Aspects of the Vaginal Speculum 8
   3.3 Advantages and Disadvantages of Current Vaginal Speculums 9
4. Improved Product Specification 10
   4.1 Specifications for the Proposed Device 11
      4.1.1 Specification of Imaging Components 11
      4.1.2 Specification of Sample Collection Components 13
      4.1.3 General Specifications for Patient Comfort 14
      4.1.4 Data Protection 14
   4.2 Method of Operation 15
5. Conclusion 16
6. References 17
7. Appendix 20

1. Introduction

Cervical cancer remains a significant global health concern, with approximately 660,000 new cases and 350,000 deaths reported in 2022 [1]. This is the fourth most common cancer among women worldwide, with a disproportionate burden in low- and middle-income countries, where about 94% of cervical cancer-related deaths occur. The highest incidence rates are observed in sub-Saharan Africa, Central America, and South-East Asia, largely due to inequalities in access to preventive measures such as HPV vaccination and cervical cancer screening. Despite being 100% preventable through vaccination and regular screening,

cervical cancer continues to claim thousands of lives annually. A critical factor contributing to these statistics is the low uptake of preventive services, including Pap smear tests and cervical examinations. In England, for instance, only 65.8% of eligible women aged 25-49 took up their cervical screening examinations as of December 2023, leaving a third of this age group unscreened [2]. Similarly, in the United States, the percentage of women overdue for cervical cancer screening increased from 14% in 2005 to 23% in 2019 [3]. Many women avoid these screenings due to discomfort, privacy concerns, and fear of pain associated with the use of traditional speculums [4]. In a study conducted in Great Britain, the proportion of eligible women being screened declined from 76% in 2011 to 72% in 2017 [5]. The most frequently excuses to avoid screening include being too busy, embarrassment, fear of what the test might find, and past negative experiences with screening [5].

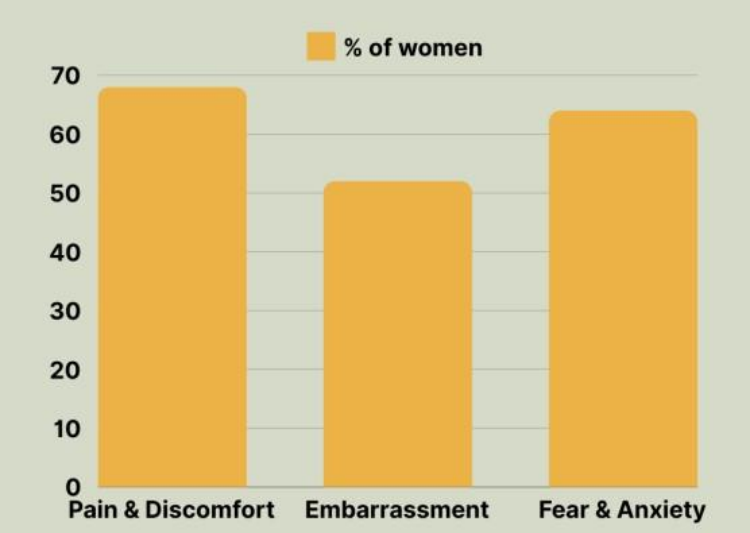

Figure 1. Study by O'Laughlin et al., 2021 showing percentages of women avoiding cervical examinations and the respective reasons

2. The Vaginal Speculum

2.1 History of the Vaginal Speculum

The vaginal speculum, the traditional device for cervical examinations, has been used in gynecological procedures over the past 20 decades with no modifications having been made to its basic design. It is used by gynecologists to open the vaginal walls in order to view and examine the cervix. The vaginal speculum is known to be the oldest medical device in the history of gynecology and obstetrics. Its origin dates back to 79AD where the device was found in remains of the volcanic eruption of Mt. Vesuvius in Pompeii [6]. Early designs of the speculum in these years was a basic structure of two blades that could manually be opened

like a corkscrew and was used to view the vaginal canal and cervix for surgical and diagnostic purposes. Dr. James Marion Sims, popularly known as the father of gynecology began experimenting surgical gynecological procedures in the mid-19th century [7]. One of the tools invented during those years of gynecological procedures was the vaginal speculum. It is quite uncertain as to which particular physician invented the current vaginal speculum as many physicians made various contributions to the design of the existing speculum. In 1845, Dr. Marion Sims started to treat slave women with fistulas (an abnormal opening that connects the vagina to the rectum, colon or bladder) [7] by surgical procedures. In viewing the cervix of the slave women, he inserted a bent spoon, which he later named the Sims Speculum, into their vagina. Marie-Ann Victoire Gillain Boivin, a French midwife, is also said to have invented another design of the vaginal speculum, which was used to dilate the vagina when screwed open. Her invention is said to have evolved into the present-day vaginal speculum.

Figure 2. Pompeii Speculum

Figure 3. Sim's Speculum

2.2 Common Materials Used in Design of Vaginal Speculum

Early speculums were crafted primarily from stainless steel [8], a durable and corrosionresistant material ideal for repeated use in clinical settings where sterilization procedures were available. However, stainless steel speculums had significant drawbacks, particularly related to patient comfort, as the metal could feel cold, and they required meticulous sterilization between uses to prevent cross-contamination. In addition, the heavy, rigid design could be intimidating and uncomfortable for patients.

To address these limitations, single-use speculums made from medical-grade plastics like polycarbonate and polystyrene were introduced [8]. These plastics are lightweight, clear for better visualization, and disposable, eliminating the need for sterilization and reducing infection risks. Today, stainless steel speculums are still used in certain clinical settings where repeated sterilization is feasible, and they are valued for their stability and precision. In

contrast, plastic speculums are preferred in outpatient and low-resource settings, where costeffectiveness and ease of disposal are priorities. Recent advancements have also introduced biodegradable polymers for single-use speculums, aligning with environmental sustainability goals without compromising functionality [9], [10].

Figure 4. Austenitic steel vaginal speculum

Figure 5. Polystyrene vaginal speculum

2.3 Patents Related to the Vaginal Speculum

In 2012, the U.S Patent Office granted patent **US-8267860-B2** [11], to the vaginal speculum designed by Comfortpat BV. This design introduced a speculum with a pair of spoon-like blades, or “spoons,” that can be inserted into the vagina. The blades are connected via a hinge pin, allowing them to open and close for ease of insertion and spreading. The unique aspect of this design lies in its operating handle and body. When combined, these features provide control over the spreading of the spoon blades, which are held at a fixed distance for a stable examination. This hinged design aims to enhance precision in blade positioning, giving clinicians better control while minimizing discomfort for patients. The ergonomic handle design further improves user control, which is valuable for prolonged or detailed examinations.

US8460187B2, patented in 2013 in the United States by Viospex [12], focuses on a conical, clear plastic or metal speculum that facilitates gentle insertion into the vaginal opening. An obturator can be used to aid in the device's insertion and expansion. Once inserted, the speculum’s unique multi-axis or radial dilation mechanism enhances visualization of the cervix. This radial dilation feature, which allows the speculum to open evenly around its periphery, offers a comprehensive view for clinicians, which is especially beneficial in detailed

examinations. Additionally, the speculum includes a handle that can accommodate a standard penlight, with a light pipe directing illumination from the light to the examination area. This design combines ease of insertion with improved lighting, creating a versatile tool for both standard and complex examinations.

Another patent, the Canadian patent **CA-2603285-C** [13] was granted in 2015 to Welch Allyn, Inc. This is by far the only patent granted to Welch Allyn Inc. after applying for patent for their design under patent **WO-2006121530-A2** [14] in five countries (Australia, China, Japan, United States, Canada), the European Patent Office, as well as the World Intellectual Property Organization (WIPO). This design under patent **CA-2603285-C** enhances the lighting and adjustability of a speculum. It includes an upper and a lower blade, with the lower blade featuring an integrated handle cavity that can receive one of two illumination assemblies. The first and second illumination options differ in size, power source, and type of light, providing flexibility in lighting configurations. This adaptability allows clinicians to choose the appropriate lighting based on the specific examination requirements, ensuring adequate visualization. The patent's emphasis on interchangeable illumination options and ergonomic blade movement supports the clinician's workflow, while the integrated lighting enhances visualization in low-light or challenging conditions.

Recently, in 2023, yet another vaginal speculum with patent number **US11547292B2** [15] was patented in the United States to Cetro Medical AB as the assignee. This modern design introduces an innovative control mechanism that enhances the usability of the speculum. The speculum includes two blade members, each extending from a proximal to a distal end. The device's defining feature is a control nut that freely spins on a threaded shaft, allowing the clinician to easily control the opening and closing of the blades. This mechanism allows for a quick and smooth adjustment of the blades, which can be crucial in a fast-paced clinical setting. The focus on ease of control and smooth blade adjustment demonstrates an evolution towards a more user-friendly design, making the speculum more accessible for both clinicians and patients by enhancing comfort and reducing the potential for abrupt or uncomfortable movements.

3. Product Analysis

The current market for vaginal speculums is diverse, with several brands leading in quality and innovation. The global market share of the vaginal speculum is projected to grow from USD 936 million in 2024 to USD 1301.5 million by the end of 2033 with a CAGR of 3.4% by the end of 2033 [16].

Figure 6. Vaginal Speculum Market (2004-2033)

3.1 Common Vaginal Speculum Brands

Becton, Dickinson and Company [17], commonly known as BD, is a major player in the medical device industry. The company has a diverse range of products, including vaginal speculums designed with a focus on safety and effectiveness. BD's speculums ensure durability while maintaining ease of use. Another key competitor in the vaginal speculum market is Medline Industries [18]. Medline Industries provides a variety of reusable and disposable speculums that cater to different healthcare settings. Their products are designed to be accessible and affordable, making them a popular choice among healthcare providers. Medline's focus on both quality and cost-effectiveness has solidified its position as a leading brand in the vaginal speculum market. CooperSurgical Inc. [19] also stands out for its innovative approach to gynecological instruments. The company has developed single-use illuminating vaginal speculums that enhance visibility during examinations. This feature is particularly beneficial for healthcare providers, as it allows for better visualization of the vaginal canal and cervix, ultimately improving diagnostic accuracy.

Welch Allyn [18] is also well-known for its reliable vaginal speculums designed with userfriendliness in mind, allowing healthcare providers to conduct examinations efficiently. Welch Allyn emphasizes ergonomic designs that reduce strain on practitioners while ensuring patient comfort during procedures. Their reputation for quality and reliability has made them a trusted name in the healthcare industry.

Yet another company involved in the manufacturing of vaginal speculums is MedGyn Products Inc. [20] which specializes in gynecological products. Their designs prioritize convenience and hygiene, making them ideal for high-volume clinical settings where infection control is important. Innovative designs like the Bouquet Speculum [21] also offer competition in the market by reducing discomfort by distributing forces symmetrically, enhancing patient comfort during examinations. There is also the Callascope [22], [23] which is a simple inexpensive product with a tube equipped with a camera to take pictures of the cervix.

Figure 7. Popular vaginal speculum brands

Figure 8. Bouquet Speculum

Figure 9. Callascope

3.2 Important Aspects of the Vaginal Speculum

Just like any other biomedical device, several essential aspects of the vaginal speculum define its functionality and clinical utility. These include biocompatibility, patient comfort, ease of use for clinicians, visualization, and infection control. The material used in speculums is a crucial factor, with options ranging from stainless steel for reusable models to plastic for disposable ones. Whether stainless steel or medical-grade plastic, the materials used must be biocompatible to avoid any adverse reactions when in contact with body tissues. Patient comfort is also a critical factor, as the exam is often uncomfortable and induces fear and anxiety. Many modern designs incorporate ergonomic handles, rounded edges, and smooth surfaces to reduce discomfort, and some brands provide multiple sizes to accommodate different patients. Typically, for functionality, vaginal speculums include mechanisms, such as screw-based or lever adjustments, to hold the device in an open position, freeing the clinician's hands for examination. Many models incorporate reflective surfaces or built-in lighting to improve visibility during examinations, ensuring better visualization of the cervix and vaginal

walls to ensure accurate diagnosis. Infection control is also a major consideration in the choice of material; disposable plastic speculums are commonly used in high-volume settings to prevent cross-contamination and minimize sterilization needs, though these come with environmental trade-offs.

3.3 Advantages and Disadvantages of Current Vaginal Speculums

The current vaginal speculums in the market have a number of advantages. Innovative designs like the Bouquet Speculum reduce discomfort by distributing forces symmetrically, enhancing patient comfort during examinations [21]. However, generally, the advantages and disadvantages of the vaginal speculum designs mainly depend on the materials used, with reusable stainless steel models and disposable plastic models each offering unique benefits [9], [10]. Stainless steel vaginal speculums are highly durable and can withstand repeated sterilization, making them cost-effective in settings where sterilization resources are available. They provide stability and precision during examinations, which is particularly beneficial in procedures requiring careful handling. Environmentally, stainless steel is a more sustainable option, as its reusability reduces waste. However, stainless steel speculums can feel cold and uncomfortable for patients, and their rigidity can be intimidating [9], [10]. Additionally, these speculums require thorough sterilization between uses, which increases time and costs in settings with high patient turnover. Stainless steel models are also relatively heavy, which may make them more challenging for clinicians to handle in long procedures and can contribute to patient anxiety.

Disposable plastic speculums offer a different set of advantages [9]. These speculums are generally more comfortable for patients due to their lightweight structure, smooth surfaces, and absence of the cold sensation associated with metal. Plastic models are also disposable, eliminating the need for sterilization and thus reducing infection risks, making them ideal for outpatient and high-volume clinical settings. The transparency of medical-grade plastic enhances visibility of the vaginal canal and cervix, improving diagnostic accuracy. Some plastic speculums even come with integrated lighting, such as LED bulbs, which is especially useful in low-light or resource-limited environments. Despite these benefits, plastic speculums also have significant drawbacks. When disposed, they can contribute to medical waste, creating environmental concerns as the majority are non-biodegradable. Plastic speculums can also be

less durable and stable because they are prone to breakage or deformation. These speculums also are not useful for patients with vaginal wall prolapse where their vaginal walls keep collapsing and closing the speculum, affecting the effectiveness of the entire examination [24], [25].

Designs like the Callascope and other optical instruments were introduced to solve the general discomfort, fear and anxiety of the vaginal speculum [23]. However. The Callascope only takes images of the cervix and does not offer the opportunity to collect samples for biopsies for further analysis where the images are not sufficient data to detect cervical cancer.

4. Improved Product Specification

Each type of vaginal speculum in the market has clear limitations, and yet, the vaginal speculum has been used in gynecological procedures over the past 20 decades with no modifications having been made to its basic design. The new design of the vaginal speculum being proposed in this project has been named TAMPULUM (derived from the tampon size of this speculum designed device) and is expected to increase privacy, alleviate pain associated with the speculum, as well as provide a wide ranged view of the cervix for more effective diagnoses with biocompatible materials to ensure fast and efficient sterilization between uses. The proposed device would be made up of various components to enable the efficient viewing of the cervix as well as an efficient collection of cervical cell samples incorporating LEDs, lenses and optical fibres to obtain internal anatomic images to enable the analysis of organs and tissues. This can be summarized in the objective tree below.

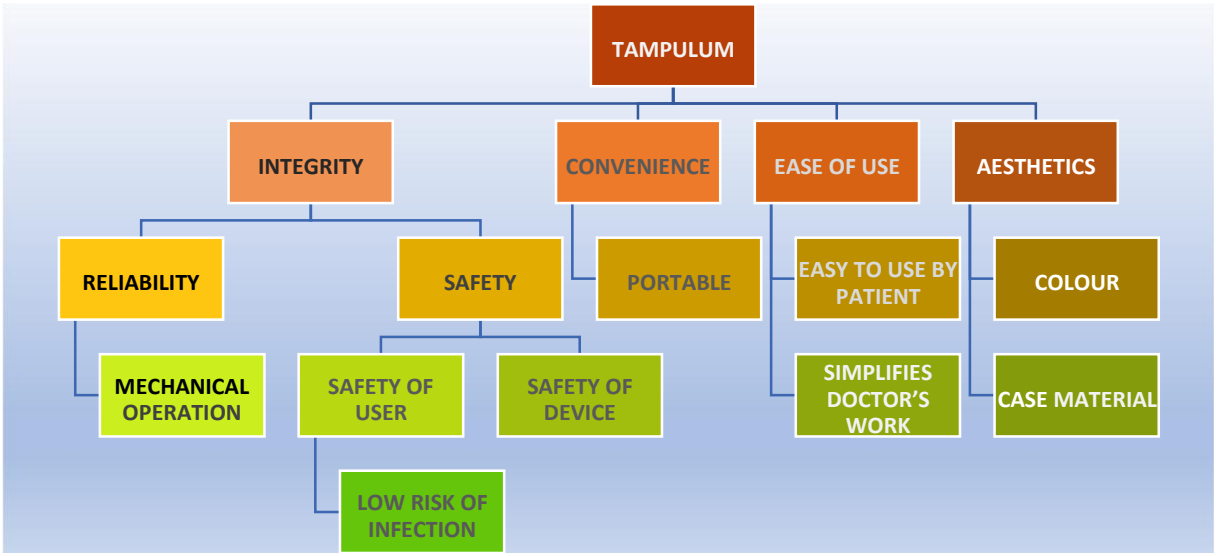

Figure 10. Objective tree of the proposed device

4.1 Specifications for the Proposed Device

4.1.1 Specification of Imaging Components

The design of the proposed device would make use of two 1/11 series miniature endoscopic lenses with 2mm outer diameter, and total length of 1.7mm each [26]. The endoscopic lens is made of medical grade epoxy which is resistant to humidity [27] and thus, makes the selected endoscopic lens the best choice for such a highly humid procedure. Also, three optic fibre cables, made up of bundles of optical fibres would utilized as a source of light transmission. Each optic fibre cable is of a 3mm diameter making it very efficient for the small diameter of the proposed device. Two durable SMD LEDs with high intensity output of about 20mA, low power consumption of about 2V, 130° wide viewing angle and 1.6mm×0.8mm×0.6mm dimensions would also be employed in the design.

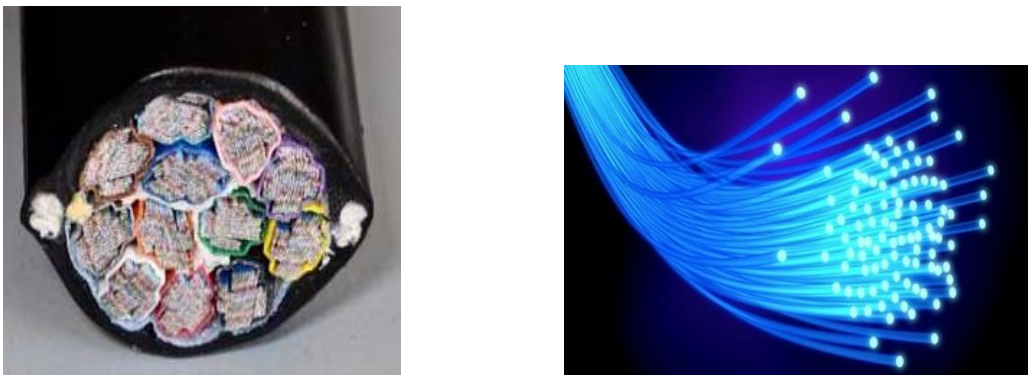

Figure 11. Optical fibers

Figure 12. 1/11 series miniature endoscopic lens with 1.77mm length and 2.0mm outer diameter

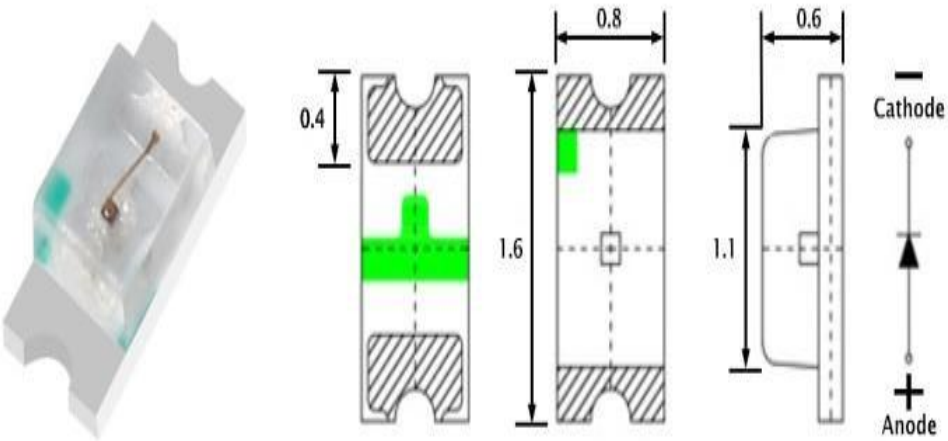

Figure 13. SMD LED

Additionally, a miniature single board computer (SBC) and a Raspberry Pi operating system would be used to ensure a successful imaging procedure. The lens is connected to the optic fibre cable, which is then connected to the viewing lens. Light emitted from the LED, already connected to a battery is sent to the object of focus and this light is reflected by the surface of the object onto the lens through total internal reflection. The light from the LED illuminates the inside of the vagina which is reflected off by the vaginal walls into the endoscopic lens. This lens collects the light reflected and transmits the reflected light through the bundle of optic fibres by the principle of total internal reflection, thereby forming an image which is displayed on the viewing lens. The SBC which is connected to the video system and the display screen runs on the Raspbian Pi operating system. To make video transmission possible, the SBC is placed on hotspot mode using RaspAP. The wireless video signal would be streamed and viewed on a standard computer monitor or projector screen using any internet browser at the doctor's workstation, outside of the main examination room.

Figure 14. Single Board Computer (SBC)

4.1.2 Specification of Sample Collection Components

For the collection of samples of cells from the cervix for biopsies or any further analysis, disposable pap smear test brush and spatula would be used. The brush and spatula are ejected from the device using a mini-pulley mechanical operating system, made up of miniature gears. This operating system is selected because it is a more reliable system as the pulley system controls the pressure exerted in the ejection of the brush and spatula. The control of this pulley system is placed on the end of the device found outside the vagina for easy use and accessibility.

Figure 15. Brush and Spatula for Sample Collection

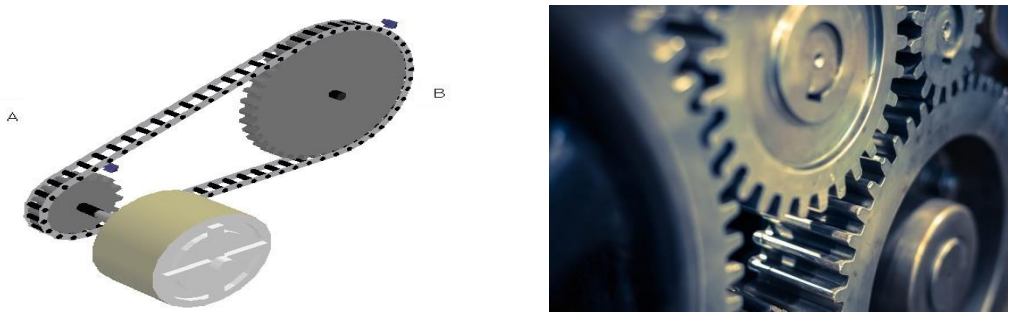

Figure 16. Pulley system of gears for mechanical operation

4.1.3 General Specifications for Patient Comfort

To ensure that the device is always warm and comfortable for its invasive use, a soft medicalgrade silicone material can be used for the case of the device. Medical-grade silicone can be sterilized and has many useful characteristics including thermal stability, low toxicity and low chemical reactivity. It also has the ability to repel water and inhibit the growth of microbes [27], [28]. This makes medical-grade silicone a very suitable case material for the device, considering its usage as it is already used in the design of menstrual cups. To allow a painless insertion of the device as well as to cater for a number of different types of female reproductive systems, the device is designed to be 155mm long and 8mm wide as shown in Figure 1, making it miniature and even smaller than the size of a mini tampon [29]. The average distance from the vaginal opening to the cervix is approximately 62.7mm to 100.27 [30] and as such, the proposed device can cater for female reproductive systems with short distances from the vaginal opening to the cervix as well as those with relatively longer distances from the vaginal opening to the cervix.

Figure 17. Medical grade silicon typically used for menstrual cup designs

4.1.4 Data Protection

To ensure robust data protection, a multi-layered security approach would be implemented using AES-256 encryption. AES-256 encryption is essential for safeguarding sensitive information, providing strong security through a 256-bit key length that is highly resistant to unauthorized access and potential breaches [31]. This symmetric key algorithm encrypts data both at rest and in transit, making it an industry standard for protecting confidential data in healthcare. In addition to AES-256, WPA3 encryption would be utilized to significantly enhance the security. WPA3 offers individualized data encryption for connected devices [32], which is

particularly beneficial for the proposed device and general Internet of Things (IoT) devices. This feature helps prevent unauthorized access and eavesdropping on communications, thus securing the WLAN connection against interception[32].

Furthermore, configuring the Single Board Computer (SBC) with HTTPS via SSL/TLS would ensure encrypted communication between devices and viewing platforms, further mitigating risks associated with data streaming [33]. Also, to prevent unauthorized network connections, organizations should implement firewalls and Virtual Private Networks (VPNs). These tools would help monitor and control incoming and outgoing network traffic based on predetermined security rules, thus enhancing overall network security. Lastly, compliance with relevant regulations such as the Health Insurance Portability and Accountability Act (HIPAA) is crucial for data storage and access logs, ensuring that sensitive medical data is adequately protected [34], [35], [36].

4.2 Method of Operation

In order to operate the device, the patient turns on the device and inserts it into her vagina. The viewing lens projects this on the display screen, connected to the SBC. Using the Raspberry Pi Operating System, the imaging produced on the screen is viewed on a computer monitor or any alternative screen via Wi-Fi. The patient can also take samples of tissues and cells from the cervix by maneuvering the pulley system to eject the pap smear brush or spatula into the cervix when necessary. The camera would also help with an effective placement of the brush and spatula for efficient sample collection. The radiologist would also have the opportunity to save the video frames for further analysis. The principle of operation of the device has been simplified in the workflow below.

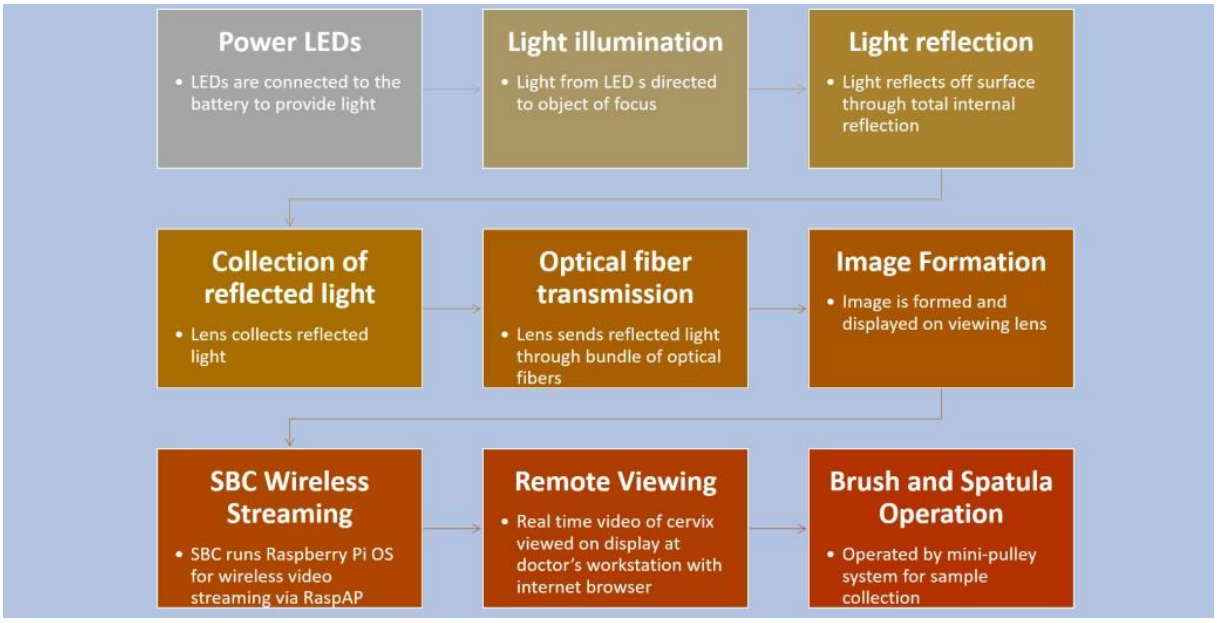

Figure 18. Principle of Operation of Device

5. Conclusion

With the improved privacy and the opportunity for people with cervixes to have control over their own bodies during cervical exams, this device has the potential to increase the frequency and regularity of cervical exams, thereby helping to detect cervical cancer, HPV, HIV, amongst many other sexually transmitted infections. It also seeks to at least decrease, an at most eradicate the notion that the vaginal speculum is a tool designed against black women. This would bridge the gap between women, especially black women, and the gynecological sector. For aesthetic purposes, the medical-grade silicone case material can be produced in a variety of colors and designs to suit the general preference of women and people with female reproductive systems. A user manual is also necessary for a successful and effective use of the device. Additionally, the device can be sterilized for reuse and is adaptable for at-home cervical imaging to increase accessibility for women in remote areas.

Future work could focus on the integration of Artificial Intelligence for real-time analysis and early detection of abnormalities, improving diagnostic capabilities. Also, it is imperative to study the cultural barriers and acceptance of this device and self-administered cervical exams in diverse communities. Finally, in order to assess the performance of the device, clinical trials are necessary.

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7. Appendix

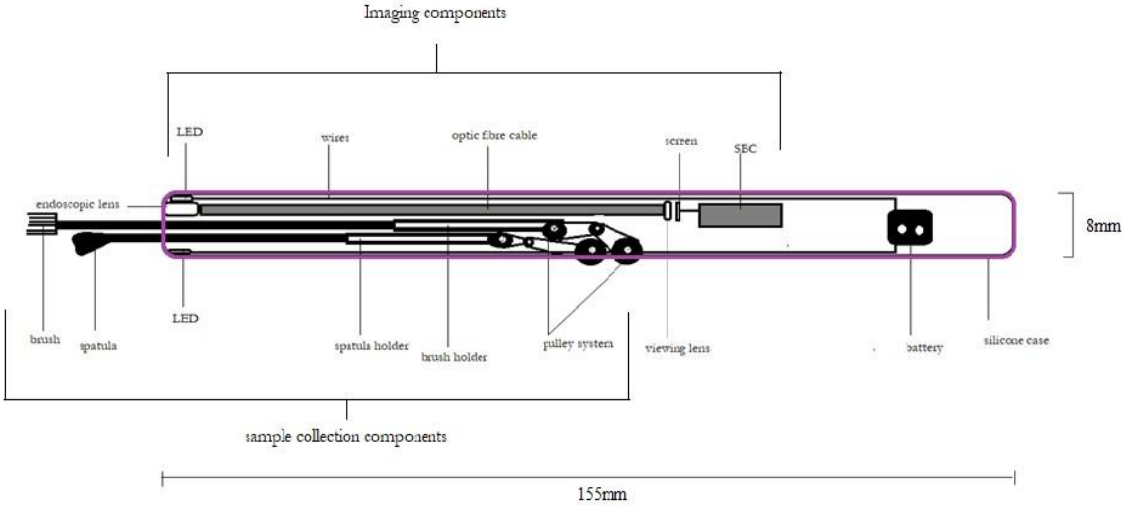

Figure 19. Parts of the proposed device with dimensions

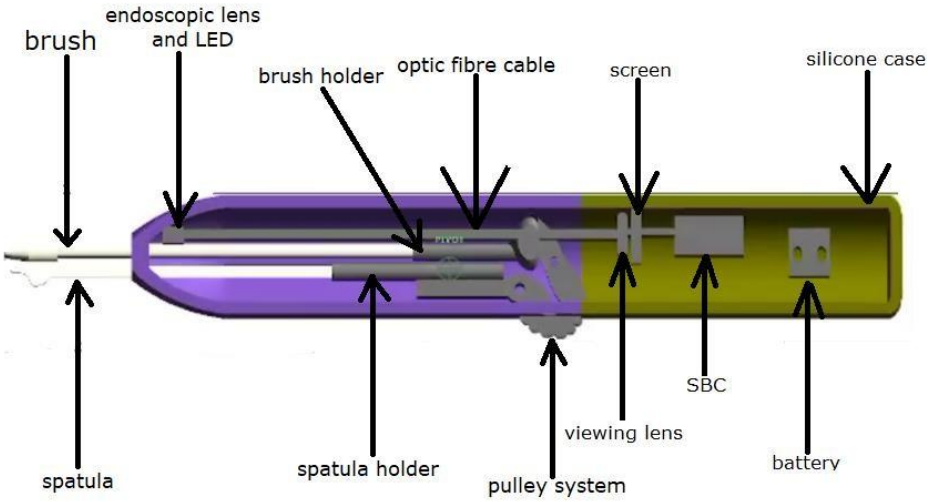

Figure 20.CAD representation showing various components of the device

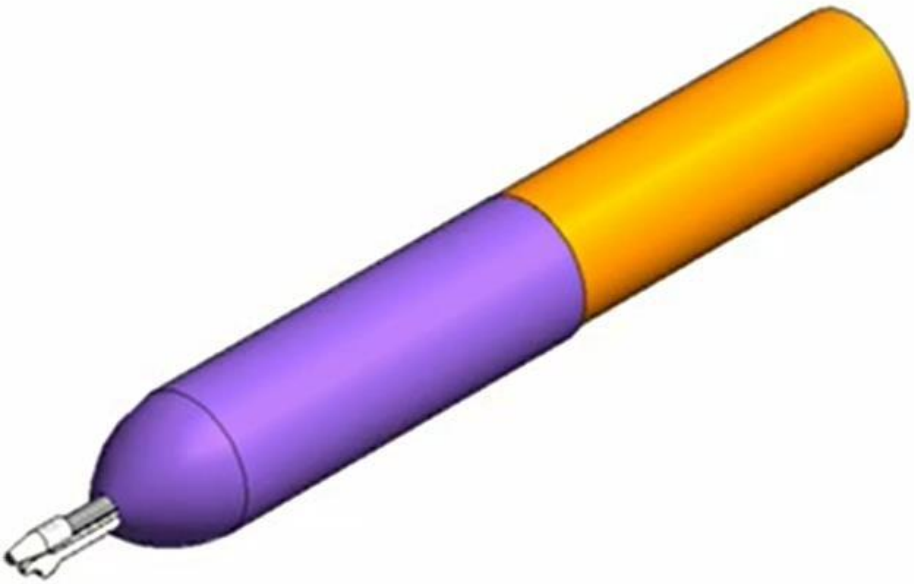

Figure 21. CAD representation of the proposed device with ejected brush and spatula